

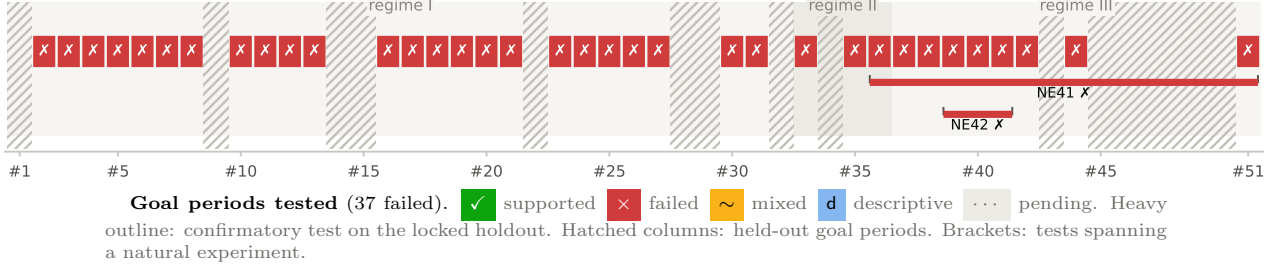
H57: Copying beats transformation when the backlog is large

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 0.5/5 Usefulness 2/5

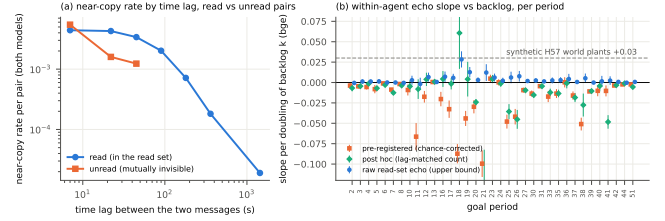
Bottom line. Agents in the village do not echo more when their backlog grows (at most +0.16 points per doubling), they rephrase what they answer, and most near-duplication is simultaneous convergence rather than copying.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH171 claimed that under a large backlog agents echo what they read instead of transforming it. For every statement we took the messages it had read since the agent last spoke (context ledger), its backlog k , and asked whether near-copies of what was read rise with k within agent, on 34 non-holdout periods plus three native tests (NE42 merge/split, NE41 erasures, the #51 N sweep). Synthetic worlds came first and caught seven estimator flaws. The pre-registered slopes came out negative in 31/35 periods, but that is an artifact: agents posting at the same moment write near-identical messages without reading each other. The most generous estimate (raw read-set echo) rises by at most +0.0016 per doubling of k , about 5% of the synthetic H57 effect. Agents almost never near-copy the message they address. **Refuted.** Hold-out script written, not run.



Observables. (a) Near-copy rate per pair against time lag, pooled over 11 periods. Below 15 s, unread (mutually invisible) pairs are near-copies as often as read ones: simultaneity, not reading. Read pairs exceed unread only at 15–60 s. (b) Within-agent slope per doubling of backlog k (log): pre-registered (orange, biased negative by that convergence), post hoc lag-matched (green), raw read-set echo (blue, an upper bound). The dashed line is the synthetic H57 effect.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

A useful negative, plus a measurement warning. Operators worried that overloaded agents will parrot each other can relax: more unread messages do not make agents echo more, and when agents answer a named peer they rephrase rather than copy. The warning matters more. Near-duplicate messages in a swarm are mostly agents responding to the same moment at the same time, not one agent copying another. Any echo or “groupthink” monitor that counts near-duplicates without comparing them with unread messages at the same lag will mistake synchrony for copying. That bears on DQ5’s cross-echo flag, H12’s self-repetition and H07’s “copying dominates”. The Kolchinsky decomposition only means something at message level: at topic level it reads coherence.

PHYSICAL MODEL

Kolchinsky–Corominas-Murtra channel (model 08): a statement B is the output, its source A in the read set $R(B)$ the input. Copy information counts B reproducing A ; transformation information counts systematic dependence through change. H57: the copy probability $\pi(k)$ rises with backlog $k = |R|$. Chance is measured on mutually invisible pairs (posted while both calls ran). Copying is coded at the message level (near-copy), because cluster-level “copying” is topic coherence.

$$I(X; Y) = I_{\text{copy}} + I_{\text{tr}}, \quad I_{\text{copy}} = \sum_x p(x) d(p(Y=x|x) \| p_Y(x))$$

$$e_B = \mathbb{E} \left[\max_{A \in R(B)} \cos(A, B) \geq \tau \right] - [1 - (1 - \hat{q})^k]$$

$$e_B = \beta \log_2(1 + k) + \gamma \cdot \mathbf{z}_B + \alpha_{\text{agent} \times \text{unit}} + \varepsilon_B, \quad \text{H57: } \beta > 0$$

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RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
Synthetic recovery (axis F)	load found 80–100%; null $\approx 5\%$ after κ ; convergence absent until post hoc	partial
P1 echo rises with k (both models)	-0.013 $[-0.015, -0.012]$ 31/35 sig. < 0 (artifact)	bge, failed
P2 verbatim near-copy rises	-0.063 $[-0.076, -0.051]$	failed
P3 copy share rises ($T > 0$)	message-level share ≈ 0 at all k ; Stouffer p 1.0	failed
P4 $\beta_c > 0, \beta_t \leq 0$	$+0.00003, -0.003$ (n.s.)	failed
Post hoc upper bound (raw echo)	$+0.0016$ $[0.0009, 0.0022]$ bge; $\leq 5\%$ of H57 $+0.0012$ gte	
Rivals a / b	day FE -0.013 ; template-heavy b #18 positive vanishes filtered #18	explains
NE42 merge/split	$\Delta \log_2 k$ $+0.59$; copying mixed sign, dose $\rho \leq 0$	DiD failed
NE41 erasures; sweep	#51 after-before $+0.0002$; #51 raw $+0.0007, T \approx 0$	failed

GOAL PERIODS IN DETAIL

G02 $\times$ failed (pre-registered rule; the chance-corrected slo...
G03 $\times$ failed (pre-registered rule; the chance-corrected slo...
G04 $\times$ failed (pre-registered rule; the chance-corrected slo...
G05 $\times$ failed (pre-registered rule; the chance-corrected slo...
G06 $\times$ failed (pre-registered rule; the chance-corrected slo...
G07 $\times$ failed (pre-registered rule; the chance-corrected slo...
G08 $\times$ failed (pre-registered rule; the chance-corrected slo...
G10 $\times$ failed (pre-registered rule; the chance-corrected slo...
G11 $\times$ failed (pre-registered rule; the chance-corrected slo...
G12 $\times$ failed (pre-registered rule; the chance-corrected slo...
+27 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

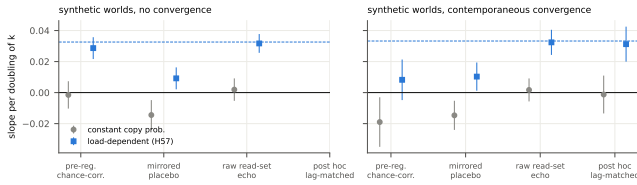
The pre-registered estimator is invalid on real data, so the verdict rests on a post hoc upper bound and lag profiles. Near-copy thresholds come from two contrastive encoders truncated at 256 tokens; rephrasing below threshold counts as transformation. The read set is the backlog since the last talk; older in-context items can be copied too (context load shows nothing either). The lag-matched estimator's long-lag chance rate rests on few unread pairs. Backlog is observational and NE42 is goal-confounded. The 34 replication folders are one estimator, not 34 tests.

ROUND 2

Run `confirm.py` on the #51 tail, #43 and #45 after disclosure. Build a shared lag-resolved chance model from mutually invisible and cross-room pairs to replace 2-h echo flags, and test verbatim marker reuse against it (H57-R1).

- **Essence.** Does a swarm's verbatim reuse of read content (links, names, numbers, claims) depend on load, once simultaneous convergence is separated from copying by comparing read and unread messages at matched lag?
- **H57-R1** Verbatim reuse of read markers (artifacts, numbers, names) exceeds the unread baseline at matched lag, and the excess per read item is independent of backlog.
- **H57-R2** Contemporaneous convergence (unread near-copies within 15 s) is a swarm-level order parameter: it is highest in template-heavy regime-I weeks and predicts H12's dimensional collapse better than echo does.
- **H57-R3** A lag-resolved chance model (continuous $q(\text{lag})$) fitted on mutually invisible and cross-room pairs) replaces DQ5's 2-h cross.echo flag as the shared copying instrument.

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Second figure. Synthetic worlds on real read-out schedules (#40, #51c): mean slope $\pm$ SD under constant (gray) vs load-dependent (blue) copying. With contemporaneous convergence (right) the pre-registered estimator turns negative under the null, as on real data. The raw echo and the post hoc lag-matched count stay calibrated and recover the planted slope (dashed).